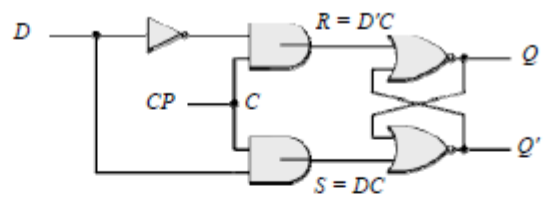
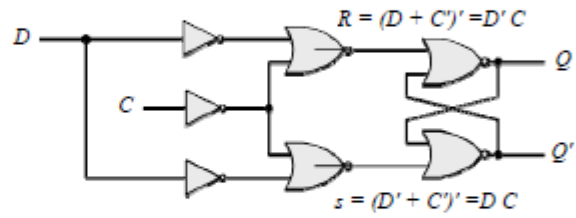


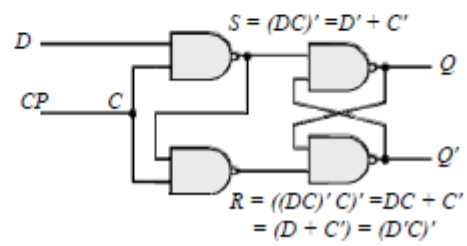
5.1 (a)



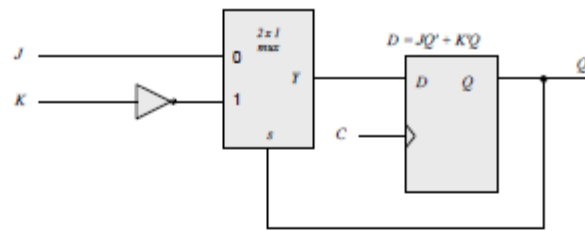
(b)



(c)



5.2



5.3 $Q'(t+1) = (TQ' + T'Q)' = (TQ')' \cdot (T'Q) = (T' + Q)(T + Q') = TQ + T'Q'$

5.4 (a) Characteristic Table:

A	B	$Q(t+1)$
0	0	$Q'(t)$
0	1	1
1	0	0
1	1	$Q(t)$

(b) Excitation Table

$Q(t)$	$Q(t+1)$	A	B
0	0	1	X
0	1	0	X
1	0	X	0
1	1	X	1

- (c) Derivation of characteristic equation:

A	B	$Q(t)$	$Q(t+1)$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

B

		$BQ(t)$			
		00	01	11	10
A	0	m_0 1	m_1 0	m_3 1	m_2 1
	1	m_4 0	m_5 0	m_7 1	m_6 0

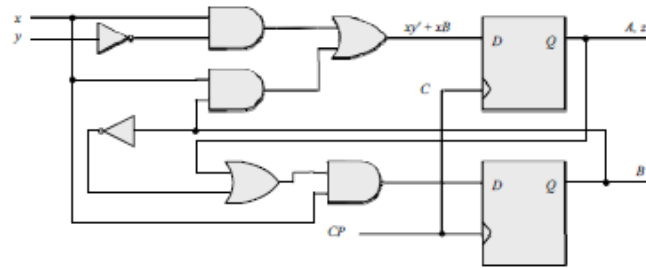
$Q(t)$

$$Q(t+1) = A'Q' + BQ$$

- (d) Connect A and B together and apply complement of T input to it.

- 5.5 The truth table describes a combinational circuit.
 The state table describes a sequential circuit.
 The characteristic table describes the operation of a flip-flop.
 The excitation table gives the values of flip-flop inputs for a given state transition.
 The four equations correspond to the algebraic expression of the four tables.

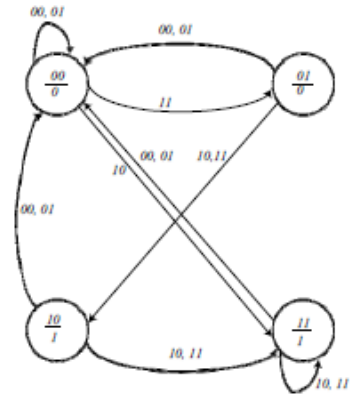
5.6



(b) $A(t+1) = xy' + xB$
 $B(t+1) = xA + xB'$
 $z = A$

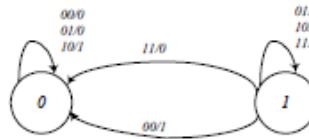
Present state		Inputs		Next state		Outputs	
A	B	x	y	A	B	z	
0	0	0	0	0	0	0	
0	0	0	1	0	0	0	
0	0	1	0	1	1	0	
0	0	1	1	0	1	0	
0	1	0	0	0	0	0	
0	1	0	1	0	0	0	
0	1	1	0	1	0	0	
0	1	1	1	1	0	0	
1	0	0	0	0	0	1	
1	0	0	1	0	0	1	
1	0	1	0	1	1	1	
1	0	1	1	1	1	1	
1	1	0	0	0	0	1	
1	1	0	1	0	0	1	
1	1	1	0	1	1	1	
1	1	1	1	1	1	1	

(c)



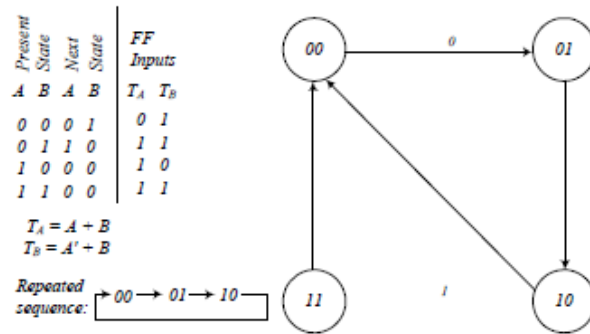
5.7

Present state	Inputs		Next state	Output
Q	x	y	Q	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



$S = x \oplus y \oplus Q$
 $Q(t+1) = xy + xQ + yQ$

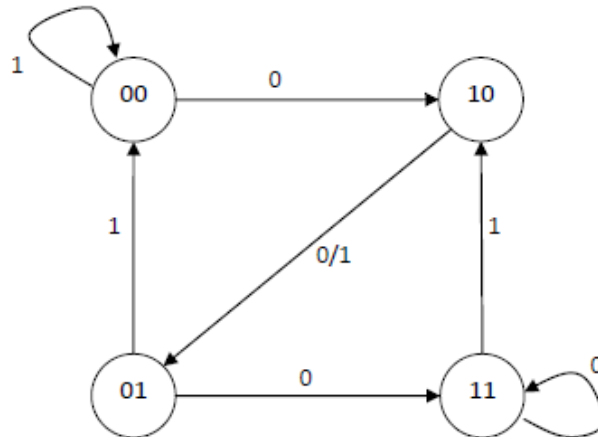
5.8 A counter with a repeated sequence of 00, 01, 10.



5.9 (a) $J_A = x'$, $K_A = B'$; $A(t+1) = JA' + K'A = x'A' + BA$
 $J_B = A$, $K_B = x$; $B(t+1) = JB' + K'B = AB' + x'B$

(b)

Present State		Input	Next state	
$A(t)$	$B(t)$	x	$A(t+1)$	$B(t+1)$
0	0	0	1	0
0	0	1	0	0
0	1	0	1	1
0	1	1	0	0
1	0	0	0	1
1	0	1	0	1
1	1	0	1	1
1	1	1	1	0



5.10

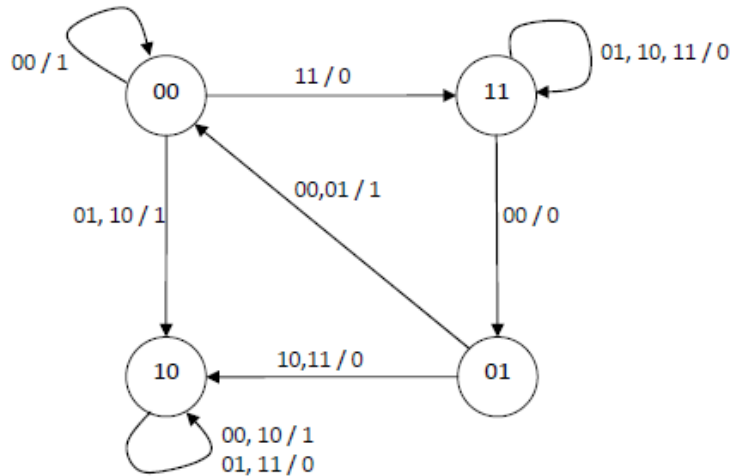
$$\begin{aligned}
 J_A &= A'x + B'y; \quad K_A = Bx'y' \\
 A(t+1) &= JA' + K'A = (A'x + B'y)A' + (Bx'y')'A = A'x + A'B'y + (B' + x + y)A \\
 &= A'x + A'B'y + AB' + Ax + Ay = x(A' + A) + B'(A + A'y) + Ay \\
 &= x + B'(A + y) + Ay = x + AB' + B'y + Ay
 \end{aligned}$$

$$B(t+1) = JB' + K'B = A'B'xy + A(B + x')B = A'B'xy + AB$$

(a) Tabulate the state table.

Present State		Input		Next state		Output	Flip – Flop Inputs			
$A(t)$	$B(t)$	x	y	$A(t+1)$	$B(t+1)$	z	J_A	K_A	J_B	K_B
0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	1	0	1	1	0	0	1
0	0	1	0	1	0	1	1	0	0	1
0	0	1	1	1	1	0	1	0	1	1
0	1	0	0	0	0	1	0	1	0	1
0	1	0	1	0	0	1	0	0	0	1
0	1	1	0	1	0	0	1	0	0	1
0	1	1	1	1	0	0	1	0	1	1
1	0	0	0	1	0	1	0	0	0	0
1	0	0	1	1	0	0	1	0	0	0
1	0	1	0	1	0	1	0	0	0	1
1	0	1	1	1	0	0	1	0	0	1
1	1	0	0	0	1	0	0	1	0	0
1	1	0	1	1	1	0	0	0	0	0
1	1	1	0	1	1	0	0	0	0	0
1	1	1	1	1	1	0	0	0	0	0

(b) Draw the state diagram of the circuit.



5.11 (a)

Present State	00	01	11	00	00	01	00	01	00	00	01	11	10	00	01
Input	1	1	0	0	1	0	1	0	0	1	1	1	0	1	0
Output	0	0	1	0	0	1	0	1	0	0	0	0	1	0	1
Next State	01	11	00	00	01	00	01	00	00	01	11	10	00	01	00

Output Sequence: 001 001 010 000 101

(b) Let $a : 00, b : 01, c : 10, d : 11$

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	a	d	1	0
c	a	c	1	0
d	a	c	1	0

We find that c is equivalent to d . So after merging c and d as c and then replacing all transitions to d by transitions to c , we get:

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	a	c	1	0
c	a	c	1	0

We find that b is equivalent to c . So after merging b and c as b and then replacing all transitions to c by transitions to b , we get:

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
a	a	b	0	0
b	a	b	1	0

This is the final reduced state table, and so we represent it in the format of Table 5.3

Present State		Next State				Output	
		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	0	1	1	0

(c)

Present State	00	01	01	00	00	01	00	01	00	00	01	01	01	00	01
Input	1	1	0	0	1	0	1	0	0	1	1	1	0	1	0
Output	0	0	1	0	0	1	0	1	0	0	0	0	1	0	1
Next State	01	01	00	00	01	00	01	00	00	01	01	01	00	01	00

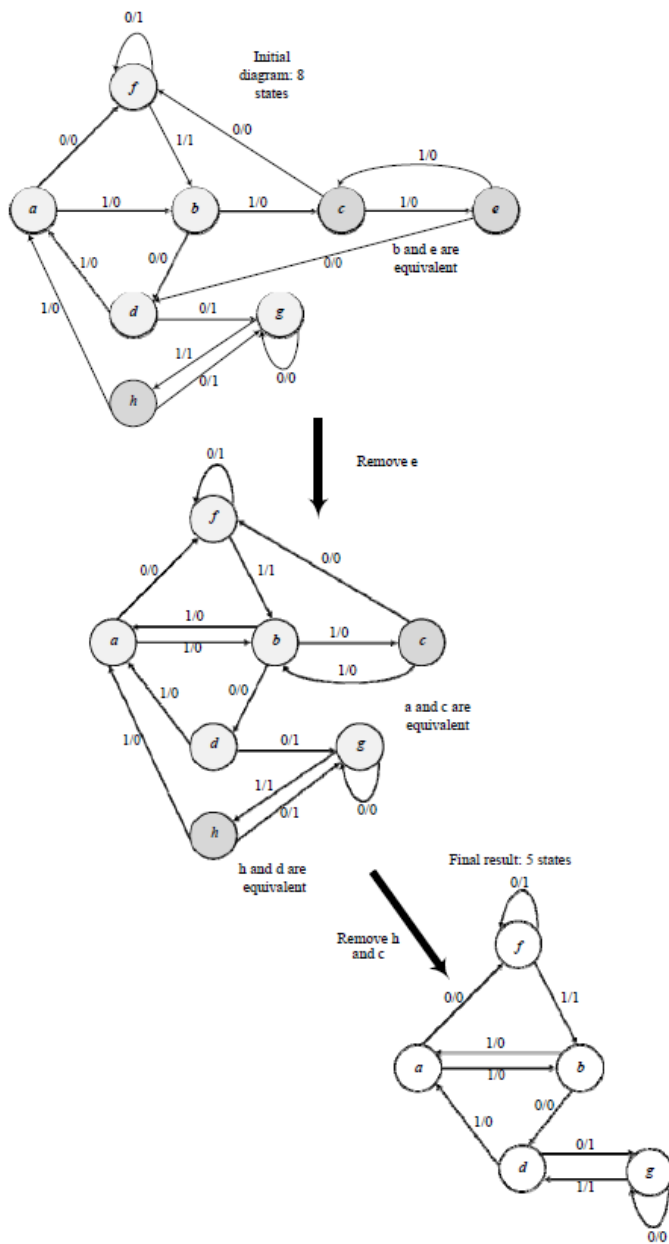
Output Sequence: 001 001 010 000 101

This is same as obtained in (a)

5.12

Present state	Next state		Output	
	0	1	0	1
<i>a</i>	<i>f</i>	<i>b</i>	0	0
<i>b</i>	<i>d</i>	<i>a</i>	0	0
<i>d</i>	<i>g</i>	<i>a</i>	1	0
<i>f</i>	<i>f</i>	<i>b</i>	1	1
<i>g</i>	<i>g</i>	<i>d</i>	0	1

Note: Equivalent states: $b = e$, $a = c$, $h = d$



5.13 Equivalent

(a) State: $a f b c e d g h g g h a$
 Input: $0 1 1 1 0 0 1 0 0 1 1$
 Output: $0 1 0 0 0 1 1 1 0 1 0$

(b) State: $a f b a b d g d g g d a$
 Input: $0 1 1 1 0 0 1 0 0 1 1$
 Output: $0 1 0 0 0 1 1 1 0 1 0$
 Equivalent states: $b = e, a = c, h = d$

5.14

Present State	Next State		Output	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
000	000	001	0	0
001	011	010	0	0
011	000	010	0	0
010	110	010	0	1
110	000	010	0	1

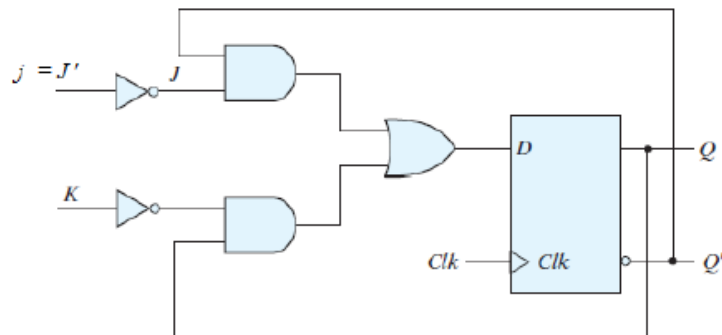
5.15

Present State	External Inputs		Inputs to internal JK flip-flop		Next State	
Q(t)	j=J'	K	J	K	Q(t+1)	
0	0	0	1	0	1	Set to 1
0	0	1	1	1	1	Complement
0	1	0	0	0	0	No change
0	1	1	0	1	0	Reset to 0
1	0	0	1	0	1	Set to 1
1	0	1	1	1	0	Complement
1	1	0	0	0	1	No change
1	1	1	0	1	0	Reset to 0

		j			
Q \ jK		00	01	11	10
	0	m_0 1	m_1 1	m_2 0	m_3 0
1	1	m_4 1	m_5 0	m_7 0	m_6 1
		K			

Characteristics Equation:

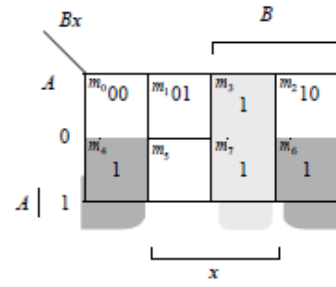
$$Q(t+1) = j'Q' + K'Q$$



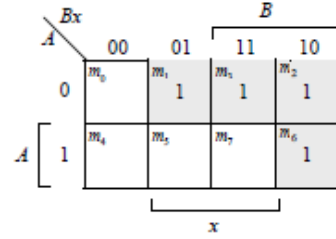
5.16 (a) $D_A = Ax' + Bx$

$$D_B = A'x + Bx'$$

Present state		Input x	Next state	
A	B		A	B
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	0



$$D_A = Ax' + Bx$$

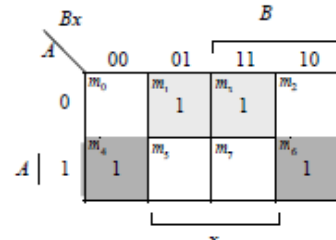


$$D_B = A'x + Bx'$$

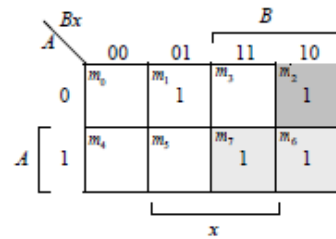
(b) $D_A = A'x + Ax'$

$$D_B = AB + Bx'$$

Present state		Input x	Next state	
A	B		A	B
0	0	0	0	0
0	0	1	1	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	1

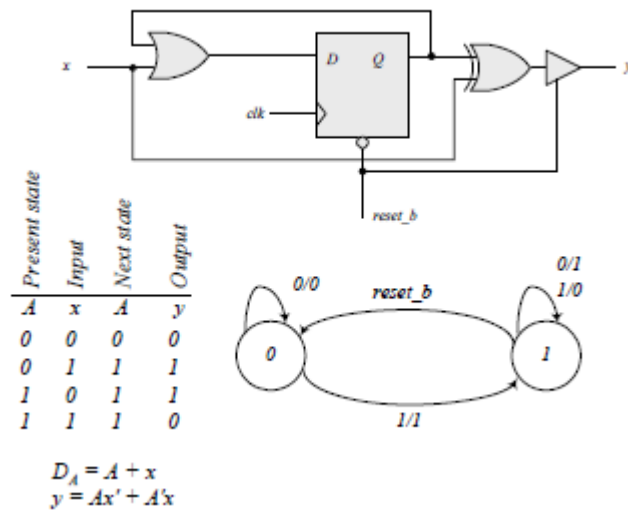


$$D_A = A'x + Ax'$$



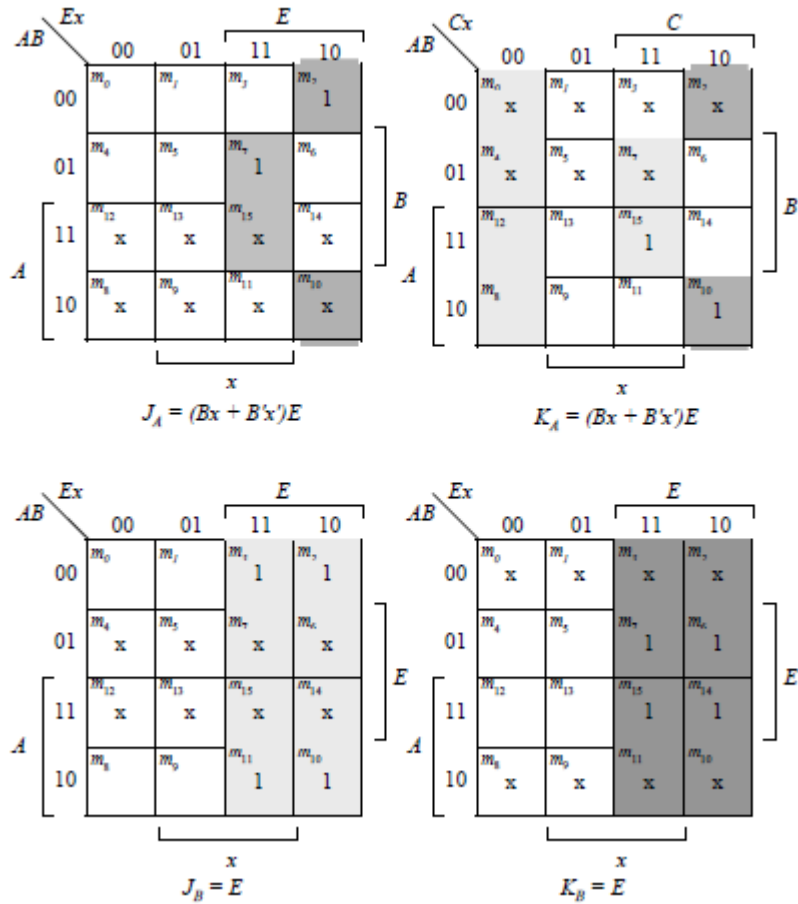
$$D_B = AB + Bx'$$

- 5.17 The output is 0 for all 0 inputs until the first 1 occurs, at which time the output is 1. Thereafter, the output is the complement of the input. The state diagram has two states. In state 0: output = input; in state 1: output = input'.



5.18 Binary up-down counter with enable E .

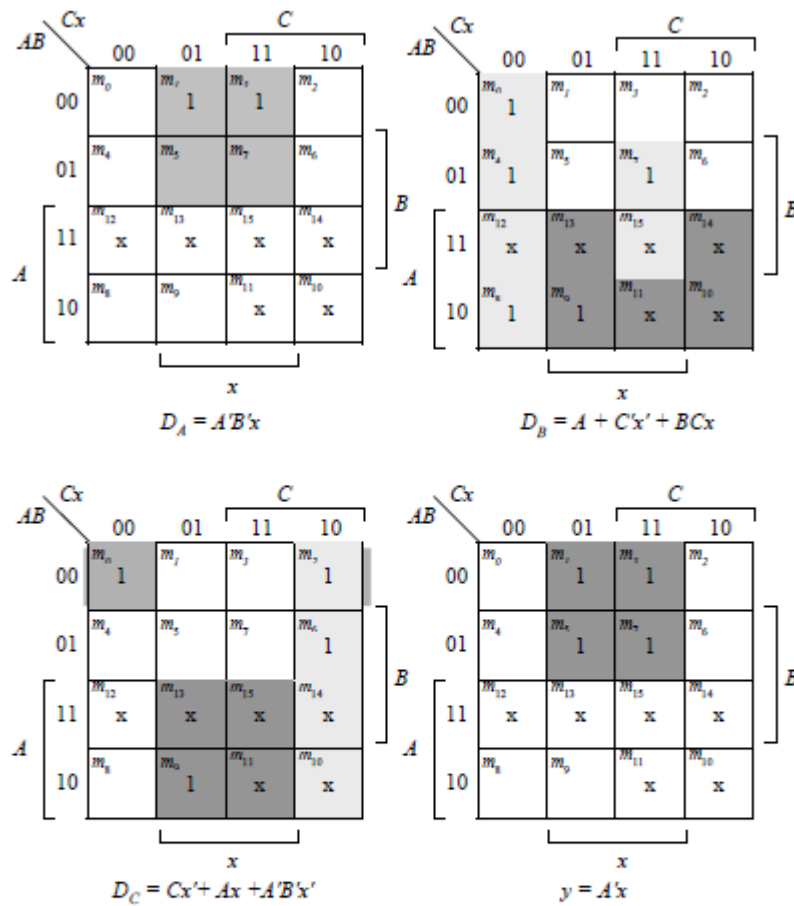
Present state AB	Input x	Next state AB	Flip-flop inputs			
			J_A	K_A	J_B	K_B
00	01	00	0	x	0	x
00	01	00	0	x	0	x
00	10	11	1	x	1	x
00	11	01	0	x	1	x
01	00	01	0	x	x	0
01	01	01	0	x	x	0
01	10	01	0	x	x	1
01	11	10	1	x	x	1
10	00	10	x	0	1	0
10	01	10	x	0	1	0
10	10	01	x	1	x	1
10	11	11	x	0	x	1
11	00	11	x	0	x	0
11	01	11	x	0	x	0
11	10	11	1	0	x	1
11	11	11	x	1	x	1



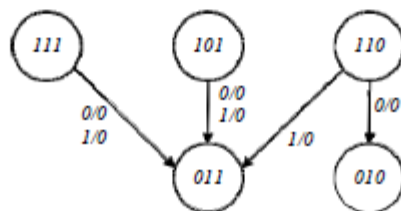
5.19 (a) Unused states (see Fig. P5.19): 101, 110, 111.

Present state <i>ABC</i>	Input <i>x</i>	Next state <i>ABC</i>	Output <i>y</i>
000	0	011	0
000	1	100	1
001	0	001	0
001	1	100	1
010	0	010	0
010	1	000	1
011	0	001	0
011	1	010	1
100	0	010	0
100	1	011	1

$$d(A, B, C, x) = \Sigma (10, 11, 12, 13, 14, 15)$$



The machine is self-correcting, i.e., the unused states transition to known states.



(b) With JK flip=flops, the state table is the same as in (a).

Flip-flop inputs					
J_A	K_A	J_B	K_B	J_C	K_C
0	x	1	x	1	x
1	x	0	x	0	x
0	x	0	x	x	0
1	x	0	x	x	1
0	x	x	0	0	x
0	x	x	1	0	x
0	x	x	1	x	0
0	x	x	0	x	1
x	1	1	x	0	x
x	1	1	x	1	x

$J_A = B'x$
 $J_B = A + C'x'$
 $J_C = Ax + A'B'x'$
 $y = A'x$

$K_A = 1$
 $K_B = C'x + Cx'$
 $K_C = x$

The machine is self-correcting because $K_A = 1$.

5.20 From the state table 5.4: $D_A(A, B, x) = \Sigma(2, 3, 4, 5, 7)$ and $D_B(A, B, x) = \Sigma(0, 2, 4, 7)$.

		B				
		Bx				
A	Bx	00	01	11	10	
		m_0	m_1	m_2	m_3	
0	0			1	1	$D_A = A'B + AB' + Bx$
1	1	1	1	1		
	0					
	1					

		B				
		Bx				
A	Bx	00	01	11	10	
		m_0	m_1	m_2	m_3	
0	0	1			1	$D_B = A'x' + B'x' + ABx$
1	1	1		1		
	0					
	1					

5.21 (a) The statements associated with an **initial** keyword execute once, in sequence, with the activity expiring after the last statement completes execution; the statements associated with the **always** keyword execute repeatedly, subject to timing control (e.g., #10).

(b) Assignments to variables execute sequentially and have immediate effect. Assignments to signals execute in sequence, but their effect occurs after all of the statements have been evaluated.